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Large eddy simulation of transient leading edge propagation in a turbulent lifted hydrogen jet flame

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ABSTRACT

A lifted turbulent jet flame involves partial premixing and exhibits a tribrachial flame structure at its leading edge (LE). Its propagation from the initial sparking location towards the final stabilisation height has rich physics. Large eddy simulation (LES) with flamelet-based reaction rate closure for partially premixed combustion is employed to study this propagation. The initial kernel grows radially in the rich region, and it is skewed by the oncoming flow as it is convected downstream. An LE is formed as the flame propagates radially into lean mixtures with low streamwise velocities. This LE encounters the lean limit, while a core flame continues to develop closer to the jet centreline, where the mixture reactivity and flow velocity are significantly large. Eventually, this core flame overtakes the LE in the lean mixture and upstream propagation ensues. The LE propagates mostly in the lean mixture, as the streamwise velocity in the vicinity of the jet core is high, although occasional rapid propagation is observed as the core flame encounters highly reactive mixture due to turbulence. Hence, the two flame branches are competing to form the LE. This behaviour is quite different to propagation in a lifted methane jet flame due to the increased reactivity and wider flammability limits of hydrogen. Turbulence plays a fundamental role during propagation for the formation of new upstream flame kernels, which evolve from pockets of hot reactants and fresh mixture. These kernels may be convected downstream causing the LE to recede. Eventually, the LE reaches a stationary state, and the flame root stabilises at a position where the burning mass flux is balanced by the flame normal advective mass flux. The LE does not encounter the value of the extinction dissipation rate for the mixture fraction during its evolution towards the stabilisation and at this location.

1. Introduction

Hydrogen (H_2) combustion systems are of great interest to address decarbonisation goals and legislations. Partially premixed turbulent combustion is common in practical systems and is achieved by injecting fuel through a simple jet. In a turbulent lifted jet flame, this partial premixing occurs upstream of the flame root and combustion is multi-modal [1,2]. These previous direct numerical simulation (DNS) studies showed that there are (1) a tribrachial flame structure at the flame root, (2) a highly unsteady turbulent fuel-rich premixed inner flame and (3) detached islands of turbulent non-premixed flames in the downstream region. The diverging streamlines at the leading edge may create locally laminar conditions [3]. The transient propagation of the flame leading edge (LE) from the initial sparking to the final stabilisation location is an important phenomenon to understand thoroughly for safety considerations and reliable burner operation. Previous DNS studies have demonstrated the effect of mixture stratification and turbulence on leading edge propagation in mixing layers [4] and decaying isotropic turbulence [5]. However, large eddy simulation (LES) studies on propagation in turbulent jets are scarce and are focused on different conditions, such as jets in vitiated co-flow under oxy-conditions [6], but similar conditions have been investigated in the past for a methane (CH_4) jet [7]. It is known that flammability limits and oxidation pathways influence flame topology and propagation [8]. Hence, it is unclear if the findings translate from CH_4 to H_2 fuelled lifted jet flames. Therefore, the aims of this work are to (1) describe and categorise the transient propagation of the flame LE from its spark location to the final stabilisation position in a turbulent, lifted H_2 jet flame, (2) analyse the observed modes of propagation and (3) identify similarities and differences of this propagation relative to a turbulent, lifted methane jet flame [7]. The remainder of this paper is organised

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as follows. The flame configuration, numerical framework, boundary conditions, grid and simulation details as well as the methods used for analysis are discussed in Section 2. The results are presented and discussed in Section 3. Section 4 summarises the findings.

2. Computational modelling details

A lifted turbulent hydrogen jet flame at atmospheric conditions, where fuel is injected with a bulk velocity of $U_0 = 680$ m/s through a round nozzle of diameter $d = 2$ mm into quiescent air, is considered here following the experiments by Cheng et al. [9] and Brockhinke et al. [10]. This flame has also been studied numerically using DNS [1, 2].

2.1. Large eddy simulation framework

The compressible, Favre-filtered conservation equations for mass and momentum are solved in the LES. The subgrid-scale (SGS) stress tensor is closed using the dynamic Smagorinsky model [11] and scalar fluxes are obtained with gradient transport models, as employed in previous studies [7,12,13]. The sub-grid turbulence–chemistry interaction model uses the Favre-filtered mixture fraction, $\widetilde{\xi}$, Favre-filtered reaction progress variable, $\widetilde{c}$, and their respective SGS variances $\sigma^2_{\xi,\mathrm{sgs}}$ and $\sigma^2_{c,\mathrm{sgs}}$ as control variables. The temperature, $\widetilde{T}$, is obtained using the transported Favre-filtered enthalpy as described in [7,12,13]. The filtered density, $\overline{\rho}$, is calculated using the state equation. The SGS dissipation rate in the $\sigma^2_{\xi,\mathrm{sgs}}$ equation is obtained from a linear relaxation model [14], while the SGS dissipation rate in the $\sigma^2_{c,\mathrm{sgs}}$ equation is closed as described in [15]. The reaction rate model is written as [7]

$$\overline{\dot{\omega}_c^*} = \underbrace{\overline{\rho}\int_0^1\int_0^1 \frac{\dot{\omega}_c(\eta,\zeta)}{\rho(\eta,\zeta)}\widetilde{P}(\eta,\zeta)\quad d\eta\, d\zeta}_{\overline{\dot{\omega}}_{\mathrm{fp}}} + \underbrace{\overline{\rho}\widetilde{c}\widetilde{\chi}_\xi\int_0^1 \frac{1}{\psi^{\mathrm{eq}}(\eta)}\frac{d^2\psi^{\mathrm{eq}}(\eta)}{d\eta^2}\widetilde{P}(\eta)\quad d\eta,}_{\overline{\dot{\omega}}_{\mathrm{np}}} \tag{1}$$

where η and ζ are the sample space variables for the mixture fraction and progress variable. The scalar dissipation rate (SDR) of the mixture fraction, $\widetilde{\chi}_\xi$, is obtained from its resolved and SGS contributions, $\widetilde{\chi}_\xi = \widetilde{D}(\nabla\widetilde{\xi}\cdot\nabla\widetilde{\xi}) + \widetilde{\chi}_{\xi,\mathrm{sgs}}$, where $\widetilde{\chi}_{\xi,\mathrm{sgs}} = 2(\nu_t/\Delta^2)\sigma^2_{\xi,\mathrm{sgs}}$. The contributions of premixed and non-premixed combustion modes are modelled through $\overline{\dot{\omega}}_{\mathrm{fp}}$ and $\overline{\dot{\omega}}_{\mathrm{np}}$ respectively. The density-weighted joint probability density function (JPDF) of ξ and c is approximated using two independent β-function PDFs as $\widetilde{P}(\eta,\zeta) \approx \widetilde{P}_\beta(\eta;\widetilde{\xi},\sigma^2_{\xi,\mathrm{sgs}})\widetilde{P}_\beta(\zeta;\widetilde{c},\sigma^2_{c,\mathrm{sgs}})$. The flamelet reaction rate, $\dot{\omega}_c$ in Eq. (1), is obtained by computing a series of freely propagating laminar flames using Cantera 2.6 [16]. To be consistent with the DNS study [1], air composition is taken as 78% N_2/22% O_2 by volume and the Westbrook mechanism [17] is used. This mechanism contains nine species and seventeen reactions. Flamelets are computed over the flammability range, $0.0095 \le \xi \le 0.15$. Here, the mixture fraction is based on Bilger's definition [18]. The progress variable is based on the water mass fraction, $c = Y_{\mathrm{H_2O}}/Y^{\mathrm{b}}_{\mathrm{H_2O}}(\xi)$, where the superscript 'b' refers to the value encountered in the burnt mixture. For further details on the combustion model and its validation, the reader is referred to earlier studies [7,12,13].

2.2. Numerical grid

The numerical simulation considers a cylindrical domain of length $225d$ and of constant diameter $100d$. A non-uniform structured grid with approximately 8.5 million hexahedral cells is employed. This grid is refined in the proximity of the fuel nozzle in streamwise and radial directions as well as in the shear layers of the fuel jet, where the minimum grid size is $\Delta = 60$ μm. This grid is gradually coarsened in the streamwise direction towards the domain outlet where $\Delta \approx 1.8$ mm. In the expected flame region, it is ensured that $\Delta^+ \le 2.5$, where $\Delta^+ = \Delta/\delta_{\mathrm{th,st}}$ and $\delta_{\mathrm{th,st}}$ is the thermal thickness of a laminar, premixed stoichiometric hydrogen–air flame. This grid resolves approximately 97% of the turbulent kinetic energy on average, which is significantly above the minimum value recommended for LES [19].

2.3. Boundary conditions

To accelerate numerical convergence, the cold-flow mixing field is initialised using unsteady Reynolds-averaged Navier–Stokes (RANS) simulation results, which are obtained from ANSYS Fluent (purely for convenience) using the Reynolds stress transport model and second-order schemes for spatial and temporal discretisation. The fuel jet velocity boundary conditions are obtained using an inflow turbulence generator based on the synthetic eddy method [20]. A separate three-dimensional steady RANS case with the Reynolds stress equation model, similar to a previous study [21] is undertaken with ANSYS Fluent to obtain the mean profiles of the velocity, the Reynolds stress tensor and the streamwise integral length scale, which are required for the inflow generator [22]. A coflow of air with a streamwise velocity of 1 m/s is used for the LES, while the cylindrical lateral surface and the outflow plane employ ambient pressure far field boundary conditions. These conditions are similar to those employed in a past RANS study [23].

2.4. Simulation details

The OpenFOAM v7 solver with its PIMPLE algorithm is used for pressure–velocity coupling (rhoPimpleFoam solver). Second-order spatial discretisation schemes are employed and a second order implicit Euler scheme with a fixed timestep of size $\Delta t = 5$ ns is used for time advancing the discretised equations. The simulation is conducted on ARCHER2, the UK National Supercomputer, using 1024 cores. The LES cold-flow field is developed for approximately $7\tau_{\mathrm{f}}$, where τ_{f} is the flow through time defined as the ratio of the computational domain length to the fuel jet bulk-mean velocity. A spherical pocket of diameter $D^+ = D/\delta_{\mathrm{th,st}} \approx 4.3$ with burnt mixture having approximately 85 times the minimum ignition energy for a stoichiometric hydrogen–air mixture [24] is deposited at the jet centreline for $x/d = 25$ at the start of the reacting flow LES. At this location, the mixture is rich ($\widetilde{\xi} \approx 0.078$) and $|U|/s^0_\ell(\widetilde{\xi}) \approx 27.7$, $\langle|U'|\rangle/s^0_\ell(\widetilde{\xi}) \approx 10.5$ and $\widetilde{\chi}^+_\xi \approx 0.008$. The unstrained freely propagating laminar flame speed, s^0_ℓ, and its thermal thickness, δ_{th}, are obtained from a Cantera flame calculation for a mixture with the same initial thermochemical state $\left(\widetilde{\xi},\widetilde{T},\overline{p}\right)$. It is known that ignition in the rich mixture is beneficial for establishing a propagating flame due to higher reaction rate and heat release rate compared to lean mixtures. Hence, heat losses are more easily overcome and early kernel development is accelerated [4]. Laminar values at stoichiometric conditions are used to normalise the reaction rate,

$$\hat{\omega} \equiv \overline{\dot{\omega}_c^*}\left(\frac{\rho_u s^0_\ell}{\delta_{\mathrm{th}}}\right)^{-1}_{\mathrm{st}}. \tag{2}$$

For the chemical kinetic mechanism used in this study, this normalisation is with 3815.28 kg/(m^{3}s). The symbol $\widetilde{\chi}^+_\xi$ is the SDR of the mixture fraction normalised by its value for extinction of a laminar non-premixed flame $\left(74\ \mathrm{s}^{-1}\right)$ [25]. The reacting flow is evolved for approximately $45\tau_{\mathrm{f}}$ which corresponds to approximately 150 laminar flame times, $\tau_\ell = (\delta_{\mathrm{th}}/s^0_\ell)_{\mathrm{st}}$. The mean field during the stabilisation phase is obtained by averaging over a timespan of approximately 5.79 ms which corresponds to $8.75\tau_{\mathrm{f}}$. The flame LE position is defined as the most upstream position of $\widetilde{T} \simeq 900$ K, following previous experimental [10] and numerical [23] studies. This position is tracked in a laboratory (Eulerian) coordinate system.

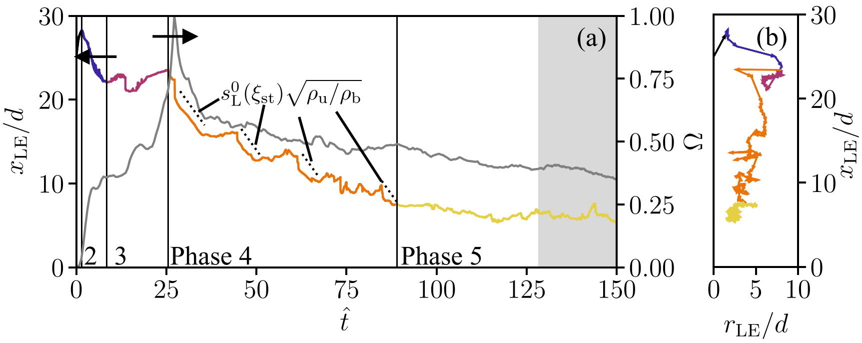

Fig. 1. (a) Transient evolution of the flame leading edge in streamwise direction, x_{LE}/d, (LHS, coloured line) and the volume-integrated reaction rate normalised by its maximum, $\Omega = \int_V \overline{\dot{\omega}_c^*} dV/(\int_V \overline{\dot{\omega}_c^*} dV)_{\max}$ (RHS, grey line). The time axis is relative to ignition and normalised by the laminar flame time, $\hat{t} = (t - t_{\mathrm{ign}})/\tau_\ell$. Also shown is the theoretical value for laminar edge flame speed [26] of a stoichiometric H_2–air mixture (dashed black lines). The shaded grey region indicates the time span for obtaining the average field at steady-state conditions. The colours of the lines correspond to the five propagation phases marked in (a). (b) Transient evolution of the normalised radial position, r_{LE}/d, of the flame leading edge.

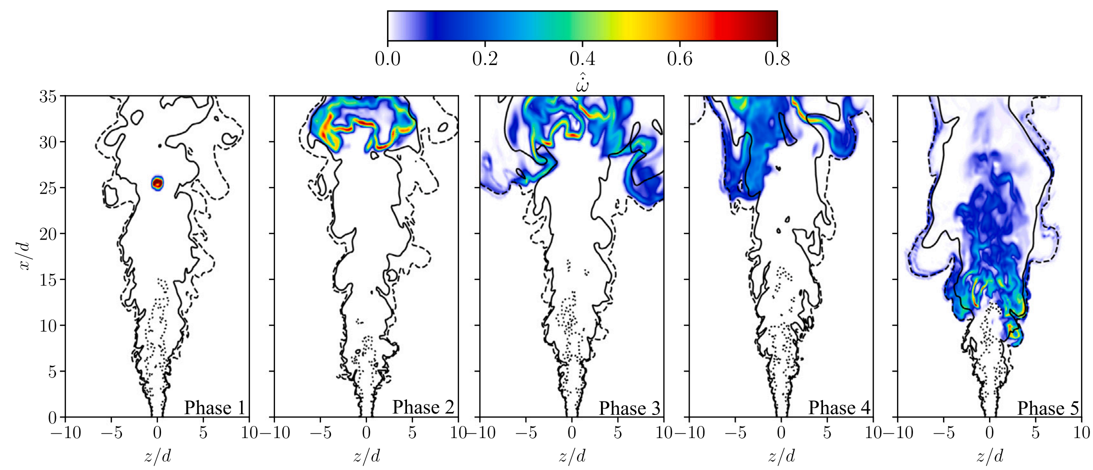

Fig. 2. Centreplane contours of the normalised instantaneous reaction rate at the beginning of each phase. Iso-lines of the stoichiometric mixture fraction, the lean and rich limits are shown using solid, dashed and dotted lines respectively.

3. Results and discussion

3.1. Analysis of flame leading edge

Five distinct phases of flame LE propagation are observed, as shown in Fig. 1: (1) ignition kernel growth and radial expansion, (2) flame leading edge formation, (3) propagation at large radius, (4) upstream propagation and (5) stabilisation. These phases are expected to last for different durations if the spark position is changed since the local velocity will be different. This affects how far the initial kernel is convected downstream during the early phases. However, the interest here is on the propagation mechanisms and not on the duration of these phases. Hence, only one spark position is considered for this study. The variation of the LE with $\hat{t}$ is shown in Fig. 1a for its streamwise position, while its radial evolution is shown in Fig. 1b. The contours of instantaneous normalised reaction rate at the time of phase transition are shown in Fig. 2. To assist the discussion on the LE dynamics, 3D animations of the 900 K iso-surface evolution are provided as supplementary material. This iso-surface is coloured by the normalised reaction rate, as shown in Eq. (2), and the timescale is relative to ignition and normalised by the laminar flame time, $\hat{t} = (t - t_{\mathrm{ign}})/\tau_\ell$.

During *Phase 1*, the initial flame kernel is convected downstream from $x/d = 25$ to 28 while it is expanding radially from $r/d = 0$ to 1.6. The streamwise flow velocity at the leading edge is positive with a mean of 25 m/s and the mean propagation speed of the LE is $\langle s_p \rangle = dx_{\mathrm{LE}}/dt = 20.13$ m/s. However, the initial kernel is not extinguished by turbulent strain. It propagates exclusively in the rich mixture ($0.037 \le \widetilde{\xi} \le 0.082$), with a mean mixture fraction of 0.057, which is nearly twice the stoichiometric value, $\xi_{\mathrm{st}} = 0.02957$. The mean normalised reaction rate is high and similar to the theoretical laminar stoichiometric flame value, $\langle \hat{\omega} \rangle = 0.884$, while its instantaneous value decreases with $\widetilde{\xi}$ from 1.668 to 0.543 as the initial kernel expands. The propagation in Phase 1 is mostly in the second quadrant (positive y, negative z) with a mean radial flow velocity of $\langle \widetilde{V}_r \rangle = -1.7$ m/s as the kernel is convected outward. Due to turbulence, the kernel is not developed symmetrically. This is seen in Fig. 2 as transition from a kernel to a propagating flame is accomplished.

During *Phase 2*, the flame propagation shifts from rich mixtures having $\widetilde{\xi} \le 0.04$ to lean mixtures with $\widetilde{\xi} \ge 0.0096$, as it begins to move upstream in regions with low streamwise flow velocity ($\langle \widetilde{U} \rangle = -4.58$ m/s), which are encountered far from the jet centreline where the mixture is lean, as seen in Fig. 2. Consequently, there is a strong increase in r/d from 1.5 to 9 and the mean propagation speed is -7.5 m/s, which is significantly above the laminar burning velocity, $s^0_\ell(\xi_{\mathrm{st}}) = 1.865$ m/s. Similarly to Phase 1, propagation occurs in the second quadrant with radial flow velocity $\langle \widetilde{V}_r \rangle = -0.69$ m/s. Mean reaction rate decreases to $\langle \hat{\omega} \rangle = 0.442$, which is approximately half the value in Phase 1, and its instantaneous value decreases with the radius as the mixture becomes increasingly fuel-leaner.

The flame front propagates exclusively in lean mixtures and frequently encounters the lean flammability limit ($\xi_{\mathrm{l}} = 0.0095$) in *Phase 3*. This is signified by the low reaction rate in the fuel-lean regions away from the jet core as observed in Fig. 2 showing the iso-lines of ξ_1. Hence, there is little upstream propagation and the LE recedes with $\langle s_p \rangle$ = 0.68 m/s. However, the flame does propagate further radially into the second quadrant and resides far from the centreline at $\langle r/d \rangle$ = 7.19, which is the highest of all phases. Here, the mean mixture fraction is a very lean 0.0108. Consequently, the reaction rate is the weakest encountered in any of the phases with $\langle \hat{\omega} \rangle = 0.160$. In a turbulent lifted methane jet flame, local extinction has been observed [7], which is not seen for H_2 due to the wider flammability limits.

Concurrently, the flame kernel that has been convected downstream continues to develop closer to the jet core ($r/d \approx 2.5$), where rich and stoichiometric mixtures are found. Hence, the normalised reaction rate in this region increases to approximately 0.4, as observed in Fig. 2 marked Phase 3 and Phase 4. Therefore, there is a time delay until the core flame in $0 \leq r/d \leq 2.5$ becomes dominant over the outer lean flame in $r/d > 5$. This core flame is fuel-rich and connects to the lean branch forming a bibrachial structure and propagates rapidly within highly reactive mixture as the LE propagation moves into *Phase 4*. This is signified by the sudden jump in the LE radial position as observed in Fig. 1b. It is also observed that the volume-integrated reaction rate increases rapidly as shown in Fig. 1a. This is because the downstream mixture is burned off, which provides "back-support" to the core flame through radicals and heat [27]. Due to benign conditions for propagation, the LE also moves in radial direction. Hence, the LE propagates into the lean region with $\langle \widetilde{\xi} \rangle = 0.0154$ and the outer lean-premixed flame branch again determines the LE. A similar chain of events is observed on several occasions during this phase. Due to its turbulent nature, the unsteady and highly fluctuating rich-premixed core flame occasionally encounters highly reactive mixture with strong variations in the upstream mixture fraction gradient. This causes rapid upstream propagation of the LE. When this occurs, the propagation speed matches the theoretical value of a laminar edge flame [26], $s_\ell^0(\xi_{st})\sqrt{\rho_u/\rho_b}$, quite well, as is shown in Fig. 1a. It should be noted that this theoretical value is relative to the local flow velocity. No such correction is considered here to maintain consistency of the coordinate system. Consequently, the agreement with the theoretical laminar edge flame propagation speed decreases with time, as the LE resides further upstream where the flow velocity is significantly increased. The variation in mixture fraction gradient decreases as the LE moves upstream in the rapid propagation mode. Due to the simultaneous radial propagation, reactivity and the LE upstream propagation speed diminish as $\widetilde{\xi}$ decreases. Hence, propagation resumes in the lean-premixed branch. Therefore, Phase 4 is characterised by a competition between rapid propagation as the inner core flame branch encounters highly reactive mixture and lean premixed propagation of the outer flame branch due to benign conditions for its propagation. Interpreting the combustion regime for these branches is difficult due to the partially premixed nature of this case and because the LES does not resolve the smallest (Kolmogorov) scale of turbulent motion and hence it is challenging to compute a meaningful Karlovitz number (Ka). Combustion regime analysis has been conducted in the DNS study [1] and the turbulence-chemistry interaction of the rich turbulent premixed flame is classified as broken reaction zones. The LE is also seen to jump in azimuthal or streamwise directions which is due to turbulence forming new upstream flame kernels, which constitutes the third propagation mode observed in Phase 4 and is discussed in more detail later. As a consequence of turbulence, the propagation is observed in the first, second and fourth quadrants. In a methane turbulent round jet, a spiral-like transient evolution of the LE has previously been observed [7], which is not seen here. It has also been observed in the methane jet that the LE attaches to the stoichiometric iso-surface, while the flame in the hydrogen jet propagates as bibrachial premixed flame where the instantaneous LE can be far from stoichiometric mixture. These effects are due to the wider flammability limits and increased reactivity of H_2. The mean LE propagation speed during Phase 4 is −2.12 m/s, which is quite close to the laminar flame speed of a stoichiometric mixture. Compared to Phase 3, the mean normalised reaction rate is increased by approximately 50% to 0.252 but its minimum value is just 0.034 in proximity of the lean limit.

The flame front stabilises in *Phase 5*. The LE is relatively stable in its streamwise position around $x_{LE}/d = 6.45 \pm 0.59$ but fluctuates more widely in its radial position $r_{LE}/d = 2.59 \pm 0.73$, where ± indicates the standard deviation. This movement is seen in Figs. 1a and 1b. The mean propagation speed is still slightly negative at −0.29 m/s, indicating upstream creep. The LE is in good agreement with the measurements of flame lift-off height, where values of $7d$ and $9.5d$ have been observed [9,10], while a value of $5.5d$ is reported in a past DNS study [1]. Similarly to Phase 4, the LE is predominantly found in the lean mixture, where $\langle \widetilde{\xi} \rangle = 0.0178$, which is consistent with the DNS study [1]. At the LE, the mean ratio of $\widetilde{U}/s_\ell^0(\widetilde{\xi})$ is 1.26, which is close to unity and indicates laminar-like conditions at the instantaneous LE due to streamline divergence, similar to previous findings in DNS studies [1,28]. However, the standard deviation of $\widetilde{U}/s_\ell^0$ is 15.95, suggesting that the instantaneous LE moves significantly, which is due to its interaction with turbulent eddies [2,3]. Consequently, the instantaneous LE is found over a wide range of mixture fractions in the range of $0.0096 \leq \widetilde{\xi} \leq 0.0638$. For the same reason, the instantaneous LE can be found in any quadrant but most frequently resides in the third. The lean-premixed outer flame branch stabilises the unstable rich-premixed inner flame branch. The frame denoted using Phase 5 in Fig. 2 shows that the reaction rate is weak close to the lean limit while the highest reaction rate is found in the rich mixture within the stoichiometric iso-lines. Compared to Phase 4, the normalised reaction rate exhibits an increased minimum of 0.058 and maximum of 1.085 at a moderate mean reaction rate of $\langle \hat{\omega} \rangle = 0.290$.

3.2. Upstream flame kernel formation

It is observed in Fig. 1a that there are instances where the LE jumps upstream during Phase 4. This is not due to the rapid propagation mode of the inner flame, as this only coincides with a jump in radial direction seen in Fig. 1b. The discontinuous propagation mode appears similar to auto-ignition, as a new 900 K kernel is formed upstream of where the earlier LE has been found and hence the LE is observed to jump upstream. However, this effect is not of chemical nature but due to turbulence, as is demonstrated in Fig. 3. In Fig. 3a, the transient evolution of the surface area of five selected $\widetilde{T}$ iso-surfaces in a small volume of $V \approx 564\delta_{th,st}^3$ are shown for one such instance where new 900 K kernels suddenly appear upstream of the previous LE. The evolution of the flame surface area, taken as the volume integral of the reaction progress variable gradient, $\nabla\widetilde{c}$, is also shown. The corresponding surfaces are shown in physical space in Fig. 3b for selected time instances. It is observed that the evolution of the 500 K iso-surface precedes the evolution of the 900 K iso-surface and flame surface indicating that there is upstream mixing of hot reacting mixture with unburnt reactants. This is also seen in physical space, as 500 K hot reactants are pushed upstream by turbulence in finger- or bubble-like structures. These structures are observed for the lean mixture with $0.0175 \leq \widetilde{\xi} \leq 0.028$, $0.09 \leq \widetilde{c} \leq 0.29$ and $300 \leq \overline{\dot{\omega}_c^*} \leq 800$ kg/(m^{3}s) and have also been observed in a turbulent lifted methane jet [7]. Their occurrence precedes the development of new 900 K kernels, which are marked by 'I' and 'II' in Fig. 3b. These kernels appear in a narrower range of $\widetilde{\xi}$ with $0.020 \leq \widetilde{\xi} \leq 0.026$ and are more reactive with $0.30 \leq \widetilde{c} \leq 0.63$ and $620 \leq \overline{\dot{\omega}_c^*} \leq 786$ kg/(m^{3}s). These kernels continue to expand radially and may be convected downstream. Hence, the LE is observed to recede, as is shown in Fig. 1a after a jump in x_{LE}/d has occurred.

3.3. Analysis of the mean field and stabilisation mechanism

As shown in Fig. 4a, the mean field exhibits a leading edge found in the lean mixture and a rich-premixed flame is observed in the core region in between the iso-lines of ξ_{st}. Some mixture is also burned

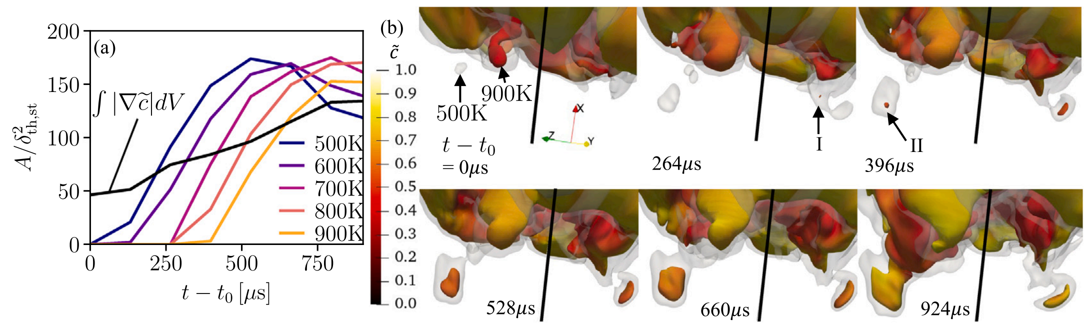

Fig. 3. (a) Transient evolution of the surface area of selected $\widetilde{T}$ iso-surfaces and flame surface area in a small volume around a new upstream $\widetilde{T}$ = 900 K kernel formation. The values are normalised by the squared thermal thickness of a stoichiometric H_2–air flame. The time is relative to an arbitrary t_0, which is the first occurrence of a 500 K kernel. (b) Iso-surfaces of 500 K (opaque) and 900 K coloured by $\widetilde{c}$ at selected instants. The jet centreline is shown in black. The appearance of the kernels is marked with 'I' and 'II'.

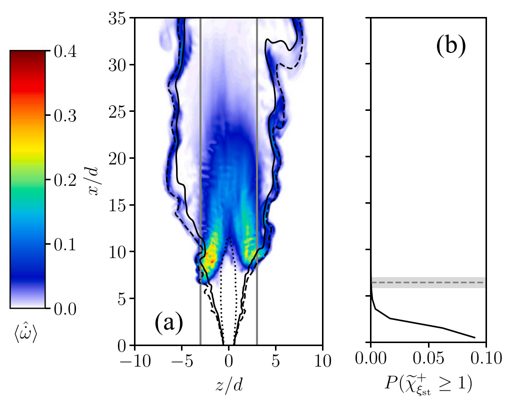

Fig. 4. (a) Centreplane contour of the normalised mean reaction rate during the stabilisation phase, indicated by the shaded region in Fig. 1a. Iso-lines of the stoichiometric mixture fraction, the lean and rich limits are shown using solid, dashed and dotted lines respectively. The vertical grey lines indicate r/d = 3. (b) Probability of encountering the extinction SDR of the mixture fraction conditioned on the iso-surface of the stoichiometric mixture fraction in streamwise direction. The lift off height is shown with a dashed grey line, and its standard deviation is shown with the grey region.

around the downstream stoichiometric surface and lean limit. These two flame branches are connected and form a bibrachial structure. Hence, the mean reaction rate exhibits the two distinctive regions of a (1) lean outer flame branch (for $r/d > 3$) and (2) rich inner flame branch (for $r/d \leq 3$) that have been developed in the propagation phase and are discussed in Section 3.1. The outer flame branch is close to the lean limit and its mean reaction rate is significantly low $\mathcal{O}(\langle\hat{\omega}\rangle) = 0.05$. Conversely, the inner flame branch that is stabilised in the high flow velocity core region has large reaction rate up to $\mathcal{O}(\langle\hat{\omega}\rangle) = 0.4$. This rich-premixed flame branch is quite unsteady but stabilised by the lean LE found in laminar-like conditions [1]. The Bunsen-like shape of the rich-premixed inner flame observed in the DNS [1] is also seen in Fig. 4a. Some asymmetry of the mean field with respect to the jet centreline is observed. Since the round jet flow is axisymmetric, this asymmetry is expected to disappear for a sufficiently long averaging window. To investigate if the flame stabilisation mechanism is of non-premixed nature, the probability of encountering $\widetilde{\chi}^{+}_{\xi_{st}} \geq 1$ is computed. This probability is obtained by using 230 instantaneous snapshots and is shown in Fig. 4b in streamwise direction. The mean position of the flame LE during the averaging window is shown with a dashed grey line, and its standard deviation is shown with the shaded region. The probability of the flame LE encountering the critical extinction value throughout flame propagation is zero and it is negligible during the stabilisation phase. Hence, the stabilisation mechanism is expected to be that of premixed flames, despite the technically non-premixed setup. This can be explained by the significant mixture stratification upstream of the flame root.

This mechanism is analysed further in Fig. 5 showing the convective, $C = (\overline{\rho}\widetilde{U})_{\perp}$, and reactive, $R = (\overline{\dot{\omega}^{*}_{c}}|\nabla\widetilde{c}_{\perp}|^{-1})$, fluxes normal to the flame leading edge, which is taken as the $\widetilde{c} = 0.05$ iso-surface to avoid the effects of non-unity Lewis numbers. The points for the scatter plot in Figs. 5a and 5b are conditioned on the radius to isolate the inner and outer flame branches. It is observed that the convective flux exceeds the reactive flux for the inner flame and that the conditional mean of their difference increases in streamwise direction. This is because of the high flow velocity encountered in the core of the near-field jet where the flame cannot stabilise. Hence, fuel is burned further downstream in a highly turbulent premixed flame and a Bunsen-flame like shape is observed, as seen in Fig. 4a, which is also seen in the DNS [1]. On the other hand, the lean outer flame branch is found in a region of lower flow velocity, where the reactive flux balances the convective flux. The scatter plot for this flame is not perfectly centred around the zero line, which is because of the limited sampling window size around the stabilisation phase shown in Fig. 1a. The scatter plot of temperature and mixture fraction coloured by the conditional mean of the reaction rate for $5 \leq x/d \leq 10$ is shown in Fig. 5c. It is observed that combustion occurs over a wide range of mixture fractions, with significant contributions of lean-premixed for $\widetilde{T} \approx 700$ K and rich-premixed, especially for $800 \leq \widetilde{T} \leq 1100$ K, despite the technically non-premixed setup. The peak temperature is encountered in the slightly rich side, as is expected due to the high fuel reactivity and heat capacity effects. The multi-modal nature of the flame root has also been observed in the DNS [1].

4. Conclusions

Partial premixing and multi-modal combustion are prevalent in practical combustion systems and the lifted jet flame is an ideal canonical candidate to study these effects. The transient propagation of a hydrogen jet flame leading edge from its initial sparking location is of interest for this study and is analysed using LES. It is observed that this propagation can be divided in several distinct phases. In the first phase, the flame kernel grows radially outwards as it is skewed and convected downstream by the incoming flow. The leading edge formed during the initial phase propagates in the lean mixtures in a region of low flow velocity with a speed significantly above the laminar burning velocity of a stoichiometric hydrogen laminar flame. This propagation continues until its speed becomes negative. This is similar to the behaviour observed in a methane jet flame [7], however, no extinction is observed for the current hydrogen flame due to its wider flammability limits and increased reactivity. In addition, a rich-premixed flame branch develops in the jet core region as it is back-supported by the downstream reactions. Due to its highly turbulent nature, this branch occasionally

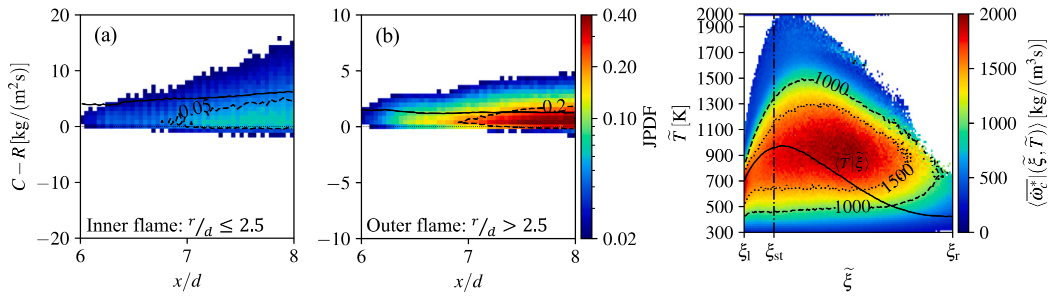

Fig. 5. Scatter plot of the difference between the convective and reacting fluxes in local flame normal direction for the (a) inner and (b) outer flames. The conditional mean is shown as the solid black line. The colour denotes the PDF. (c) Scatter plot of the temperature and mixture fraction coloured by the conditional mean of the reaction rate for the stabilisation region ($5 \leq x/d \leq 10$). The conditional mean of the temperature is shown with the solid black line.

encounters highly reactive mixture leading to rapid upstream propagation with speed similar to the theoretical value for laminar edge flames. Rapid upstream propagation coincides with slow radial propagation. Hence, there is a competition between the outer lean and inner rich flame branches for upstream propagation. These propagation modes are quite different to that observed for a methane jet flame and hence a spiral-like upstream propagation pattern is not seen in this study [7]. Turbulence also plays a fundamental role for propagation, as novel upstream flame kernels are formed through bubble- or finger-like structures of hot reactants that extend from the downstream flame. These novel kernels are observed to cause discontinuities in the streamwise leading edge position and may subsequently be convected downstream causing the leading edge to recede. These are similar to the methane flame in [7], as they are due to the flow and not of chemical nature. The flame is eventually seen to stabilise and its leading edge is found in the lean mixture where conditions are nearly laminar due to streamline divergence. The lean flame branch stabilises the inner rich-premixed flame branch. Combustion in the stabilisation region occurs throughout the entire range of flammable mixtures although it is most intense in the richer mixture. The extinction scalar dissipation rate is not encountered during the propagation of the leading edge and flame anchoring is governed by the premixed mode as flame propagation balances local flow velocity, despite the non-premixed setup. The effect of differential diffusion modelling on the flame LE propagation is part of future work and it is expected that the differential diffusion is less likely to influence the propagation mechanism discussed in this study although the duration of the various phases discussed in Fig. 1 may change.

Novelty and significance statement

Hydrogen combustion in novel burners will be in an inhomogeneous mixture within highly turbulent flows. The transient propagation of the flame leading edge (LE) initiated by a spark and evolving towards a lifted jet flame is studied using large eddy simulation employing a flamelet-based combustion closure, which allows a long observation period. The rich and lean branches of the bibrachial LE compete with one another during the propagation. This novel behaviour is different from that observed for methane jet flames due to the extended flammability limits and higher reactivity of hydrogen although the initial kernel expansion is similar for these two fuels. Flame stabilisation is in premixed mode despite the technically non-premixed configuration. These findings are novel and reported for the first time.

CRediT authorship contribution statement

Christoph D.K. Schumann: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **James C. Massey:** Writing – review & editing, Supervision, Project administration. **Caleb J. Li:** Writing – review & editing. **Nedunchezhian Swaminathan:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.proci.2025.105827.

References

[1] Y. Mizobuchi, S. Tachibana, J. Shinio, S. Ogawa, T. Takeno, A numerical analysis of the structure of a turbulent hydrogen jet lifted flame, Proc. Combust. Inst. 29 (2002) 2009–2015.
[2] Y. Mizobuchi, J. Shinjo, S. Ogawa, T. Takeno, A numerical study on the formation of diffusion flame Islands in a turbulent hydrogen jet lifted flame, Proc. Combust. Inst. 30 (2005) 611–619.
[3] A. Upatnieks, J.F. Driscoll, C.C. Rasmussen, S.L. Ceccio, Liftoff of turbulent jet flames—assessment of edge flame and other concepts using cinema-PIV, Combust. Flame 138 (2004) 259–272.
[4] N. Chakraborty, E. Mastorakos, Direct numerical simulations of localised forced ignition in turbulent mixing layers: The effects of mixture fraction and its gradient, Flow Turbul. Comb. 80 (2008) 155–186.
[5] H. Chu, L. Berger, M. Gauding, S. Pieper, H. Pitsch, Interactions of preferential diffusion and mixture inhomogeneities in hydrogen and iso-octane flame kernels under engine conditions, Comb. Flame 274 (2025) 113991.
[6] A. Rosiak, A. Tyliszczak, LES-CMC simulations of a turbulent hydrogen jet in oxy-combustion regimes, Int. J. Hydrog. Energy 41 (22) (2016) 9705–9717.
[7] Z. Chen, S. Ruan, N. Swaminathan, Large eddy simulation of flame edge evolution in a spark-ignited methane–air jet, Proc. Combust. Inst. 36 (2017) 1645–1652.
[8] F. Takahashi, V.R. Katta, Structure of propagating edge diffusion flames in hydrocarbon fuel jets, Proc. Combust. Inst. 30 (2005) 375–382.
[9] T.S. Cheng, J.A. Wehrmeyer, R.W. Pitz, Simultaneous temperature and multispecies measurement in a lifted hydrogen diffusion flame, Combust. Flame 91 (3) (1992) 323–345.
[10] A. Brockhinke, S. Haufe, K. Kohse-Höinghaus, Structural properties of lifted hydrogen jet flames measured by laser spectroscopic techniques, Combust. Flame 121 (2000) 367–377.
[11] M. Germano, U. Piomelli, P. Moin, W.H. Cabot, A dynamic subgrid-scale eddy viscosity model, Phys. Fluids 3 (7) (1991) 1760–1765.
[12] I. Langella, N. Swaminathan, Unstrained and strained flamelets for LES of premixed combustion, Combust. Theory Model. 20 (3) (2016) 410–440.
[13] J.C. Massey, Z.X. Chen., M. Stöhr, W. Meier, N. Swaminathan, On the blow-off correlation for swirl-stabilised flames with a precessing vortex core, Comb. Flame 239 (2022) 111741.

[14] C. Pierce, P. Moin, A dynamic model for subgrid-scale variance and dissipation rate of a conserved scalar, Phys. Fluids 10 (12) (1998) 3041–3044.
[15] T. Dunstan, Y. Minamoto, N. Chakraborty, N. Swaminathan, Scalar dissipation rate modelling for large eddy simulation of turbulent premixed flames, Proc. Combust. Inst. 34 (2013) 1193–1201.
[16] D.G. Goodwin, et al., 2023. available at https://www.cantera.org.
[17] C.K. Westbrook, Hydrogen oxidation kinetics in gaseous detonations, Combust. Sci. Technol. 29 (1–2) (1982) 67–81.
[18] R. Bilger, The structure of diffusion flames, Combust. Sci. Technol. 13 (1–6) (1976) 155–170.
[19] S.B. Pope, Turbulent Flows, Cambridge University Press, Cambridge, UK, 2000.
[20] N. Jarrin, S. Benhamadouche, D. Laurence, R. Prosser, A synthetic-eddy-method for generating inflow conditions for large-eddy simulations, Int. J. Heat Fluid Flow 27 (2006) 585–593.
[21] J.C. Massey, Z. Li, Z.X. Chen., Y. Tanaka, N. Swaminathan, Large eddy simulation of multi-regime combustion with a two-progress variable approach for carbon monoxide, Proc. Comb. Inst. 39 (2023) 2117–2127.
[22] N. Kornev, E. Hassel, Method of random spots for generation of synthetic inhomogeneous turbulent fields with prescribed autocorrelation functions, Commun. Numer. Methods Eng. 23 (2007) 35–43.
[23] S. Ruan, N. Swaminathan, O. Darbyshire, Modelling of turbulent lifted jet flames using flamelets: a priori assessment and a posteriori validation, Combust. Theory Model. 18 (2) (2014) 295–329.
[24] I. Glassman, Combustion, Second ed., Academic Press, London, UK, 1987.
[25] I. Kim, E. Mastorakos, Simulations of turbulent lifted jet flames with two-dimensional conditional moment closure, Proc. Combust. Inst. 30 (2005) 911–918.
[26] G.R. Ruetsch, L. Vervisch, A. Liñán, Effects of heat release on triple flames, Phys. Fluids 7 (6) (1995) 1447–1454.
[27] A.R. Masri, Partial premixing and stratification in turbulent flames, Proc. Combust. Inst. 35 (2015) 1115–1136.
[28] H.G. Im, J.H. Chen, Structure and propagation of triple flames in partially premixed hydrogen–air mixtures, Combust. Flame 119 (1999) 436–454.